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DCPR: Taming Enterprise-Scale SAP Integration API Proxies

Domain-Centric Routing Pattern & Package Domain-Centric Pattern (PDCP)

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-96% Complexity

40 → 4

Proxy Reduction

95% Faster

14.5 min

Deployment Time

100% Success

68ms

Avg Latency (33k+ Msgs)

Executive Summary

DCRP: Taming Enterprise-Scale SAP Integration

Framework Overview
2026

The Problem: Proxy Sprawl

Enterprise environments suffer from a 1:1 relationship between services and API proxies, causing chaos.

Bottlenecks	Inconsistent	Tech Debt
15 min/proxy	Security gaps	Unmanageable

The Solution: DCRP

Domain-Centric Routing Pattern: Consolidates multiple backends into a single, domain-based API Proxy using metadata-driven routing.

Engine: **JavaScript + Key Value Mapping (KVM)** eliminates infrastructure sprawl while maintaining governance.

Validated Impact

90%
Reduction in Proxies
40 → 4 consolidated

95%
Faster Deployment
273 min → 14.5 min

68ms
Avg Latency
33,000+ Msgs Tested

100%
Success Rate
Zero Downtime

My Contribution

Dynamic routing is a native feature, but tech alone isn't enough. DCRP provides the **Governance Framework** to scale it:

- Standardized Naming Conventions
- KVM Structure & Organization
- JavaScript "Brain Logic" & Templates

Target Audience

- Integration Architects
- Platform Owners
- Governance Teams
- SAP BTP Experts

Technical Prerequisites

SAP API Management
Deep knowledge of ProxyEndpoint, TargetEndpoint, Policies, and KVM configuration.

JavaScript in APIM
Context variable manipulation, error handling, JSON parsing.

Flow Architecture
PreFlow/PostFlow execution order, target.url modification logic.

Enterprise Governance
Naming conventions, change management, security policies.

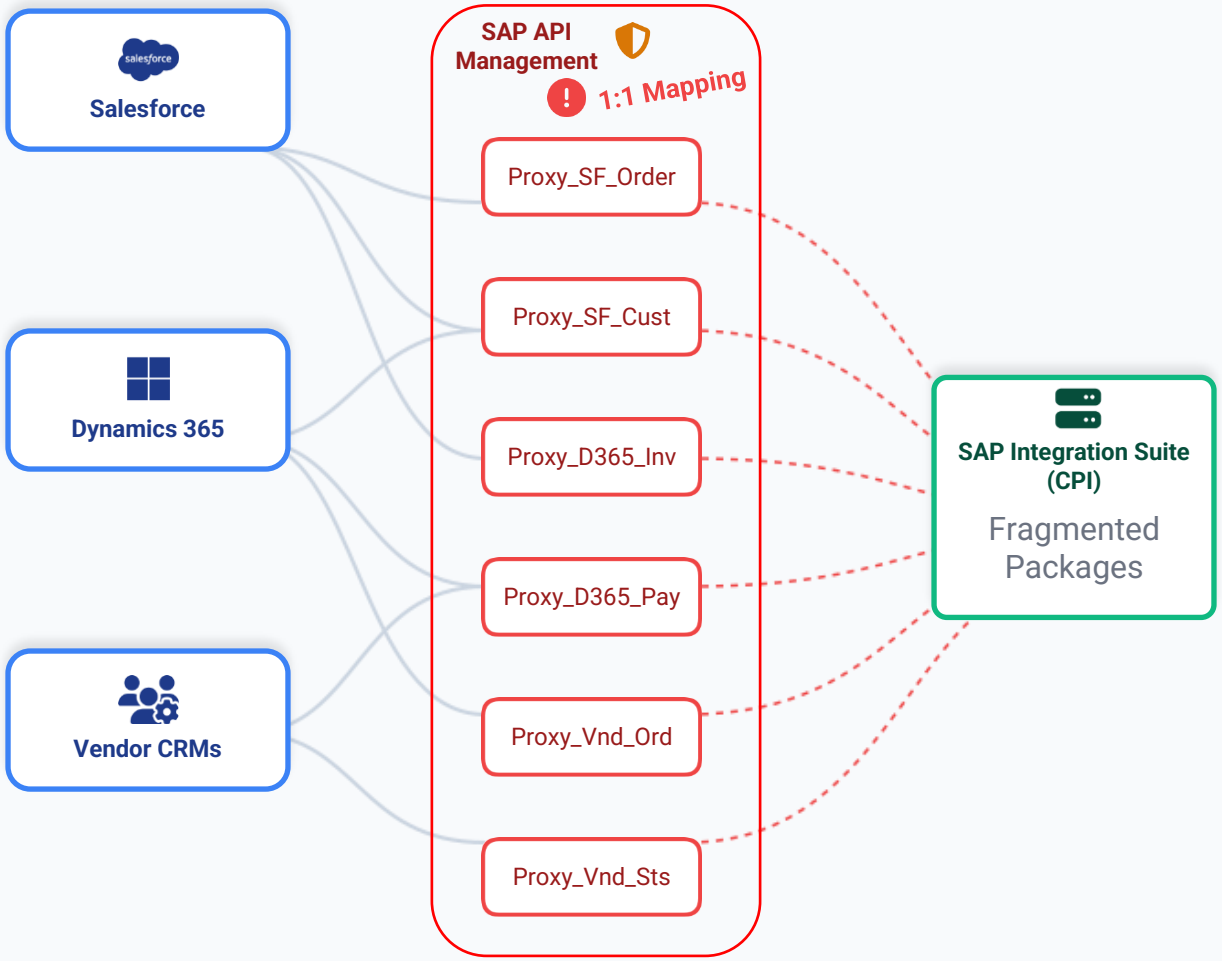
The "Proxy Sprawl" Problem

Why the Traditional 1:1 Model Fails at Enterprise Scale

Architectural Gap

Growing Tech Debt

Landscape Topology: 1:1 Chaos



Symptoms of Fragmentation

Deployment Bottlenecks
Every new integration requires a full proxy lifecycle deployment.

Inconsistent Security
Policies manually replicated across 100+ proxies lead to drift.

Fragmented Monitoring
No consolidated view of business domain traffic.

The Cost of Scale

25 Hours
Dev Time / 100 Integrations

Calculation:
100 integrations × 15 min setup

Operational Risks:

- Unacceptable Technical Debt
- Volatile SLA
- High Maintenance Costs
- Audit Failures

What is DCRP? The Three Pillars

Architecture Overview & Core Principles

Design Pattern

Foundation

> The **Domain-Centric Routing Pattern (DCRP)** is an architectural design pattern that consolidates multiple backend services into a single, business-domain-oriented API Proxy.
>
> Instead of organizing APIs into technical silos, DCRP aligns them with the organization's business domains, significantly reducing fragmentation.

Domain-Centric Organization

APIs are grouped by **Business Domain** (Sales, Finance, Logistics), NOT by technical system source.

- ✓ **Sales Domain** (SF, D365, etc.)
- ✓ **Finance Domain** (S/4, Banks)
- ✗ Salesforce Proxy

Less Fragmentation

Metadata-Driven Routing

Routing logic is decoupled from code. **KeyValueMap (KVM)** stores path mappings dynamically.

- Changes in **~30 seconds**
- No Proxy Redeployment needed
- KVM: path → backend_url

```
Idinterface:orders: <iflowname>,tracking:<iflowname>,status:<iflowname>
```

Zero Downtime **Low Maintenance**

Protocol-Agnostic Gateway

The gateway supports diverse backends seamlessly without protocol lock-in.

- Supports **REST (JSON)**
- Supports **SOAP (XML)**
- **SAP API Product** Compatible

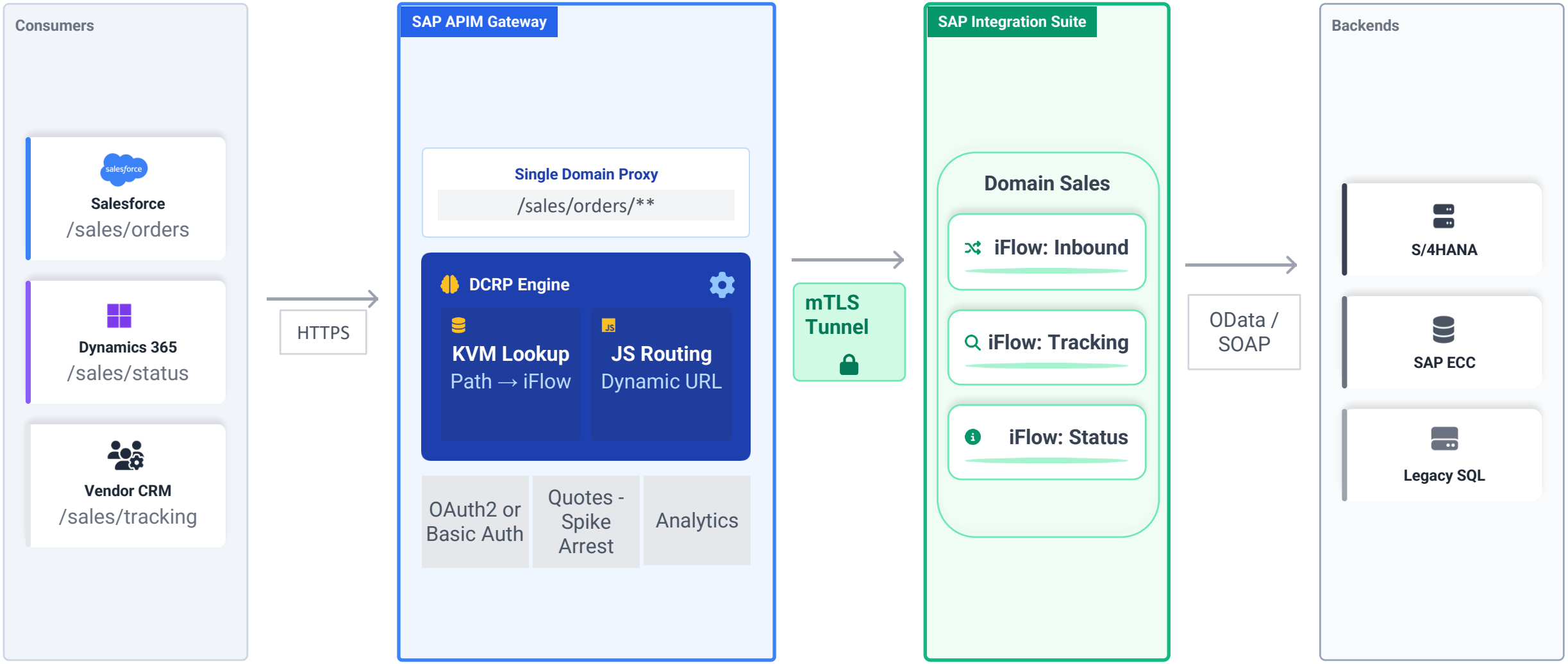
REST ↔ SOAP

Future Proof

End-to-End Architectural View

From Multi-Consumer to Unified Domain Routing

Architecture Pattern
Full Landscape



Analytics Collector Active | Capturing: X-CorrelationId, X-SenderId | Governance Layer: Centralized Security | Unified Traceability

KVM + JavaScript Foundation

The Technical Engine: "Memory" & "Brain"

Technical Deep Dive

Core Logic

The Memory (KVM)

Metadata Storage & Configuration

KVM Map Identifier:

cpipackage-nx.sales.domain.process

Key Structure:

Key	Value (Example)
packagename	nx.sales.domain
sapprocess	SAP SD & O2C
iflowname	inbound:Iflow_Inb,tracking:Iflow_Trk

Reserved Character

The colon : separates the HTTP Path from the iFlow Name. Do not use colons inside names.

Supported Delimiters

,	;	|	/	\	[
]	(	)	-	_	!

The Brain (JavaScript)

Execution Logic & Routing Engine

DCRP_Routing_Engine_v8.js

```
// 1. Fetch Metadata from "Memory"
varkvmString = context.getVariable("kvm.iflowname");
varpathSuffix = context.getVariable("proxy.pathsuffix");

// 2. Parse Logic (Split by delimiter)
varroutes = kvmString.split(","); // e.g. inbound:IflowA,tracking:IflowB
vartargetIflow = null;

// 3. Match Incoming Path to iFlow
routes.forEach(function(pair){
varparts = pair.split(":"); // [0]=path, [1]=iflow
if (pathSuffix.includes(parts[0])) {
targetIflow = parts[1];
 }
 });

// 4. Construct Final URL & Inject Headers
if (targetIflow) {
context.setVariable("target.url", "https://cpi/http/" + targetIflow);
context.setVariable("request.header.X-CorrelationId", msgId);
 } else {
throw newError("Route Not Found in KVM");
 }
```

Structured Error Handling

Governance Headers

DCRP Decision Framework

Strategic Selection Criteria: When to Apply

Strategy Guide

Evaluation Matrix

USE DCRP WHEN...

High Volume, High Value Scenarios

Domain Density

More than **10+ APIs** within the same business domain.

High Volatility

Frequent changes (monthly/weekly) requiring rapid updates.

Central Governance

Critical security & audit requirements needing uniform enforcement.

Velocity Needed

Business demands deployment times in minutes, not days.

DO NOT USE WHEN...

Low Complexity, Isolated Scenarios

Unrelated Services

Disparate APIs with no logical business connection.

Low Volume

Only 1-2 services per domain (Micro-proxy overkill).

Small Scale

Fewer than 10 total APIs in the entire landscape.

Low Maturity

Teams lack governance discipline for naming conventions.

Decision Matrix

Architect's Note: The goal is to maximize efficiency in rich, dynamic domains while avoiding "Monolithic Proxies" for simple, isolated use cases.

Optimization Strategy

Motor de Roteamento: Fluxo de Execução em 3 Estágios

Architecture of the Request Pipeline

Processing Flow

Execution Path

Stage 1: ProxyEndpoint — Pre-Flow

Incoming Request

Security Shield
OAuth2 / API Key / Whitelist or Basic Auth

Request Validation
Payload Size / Content-Type

Traffic Control
Spike Arrest / Quotas

Fail Fast
Immediate rejection (401/403/429) if any policy fails.

Stage 2: Routing Rules — Decision

Path: /sales/orders/inbound

Condition
proxy.pathsuffix MatchesPath "/orders/**"

No Match
404 Not Found
Request Dropped

Match
Forward
To TargetEndpoint

Stage 3: TargetEndpoint — Pre-Flow

DCRP Brain — JavaScript Policy

- KVM Lookup (Memory)
- Parse Logic & Routing
- Set target.url

Statistics Collector
Log custom dimensions

Raise Fault
Block if routing fails (safe fail)

To SAP CPI

Last point to modify target.url is Stage 3

Naming Convention Specification

Standardization & Architectural Context

Governance

Identity Standard

Proxy Naming Pattern

Provider		Methodology			Business Domain		Subprocess
cpi	-	dcrp	-	proxy-domain	- [Domain]	-	[Subprocess]

Example: cpi-dcrp-proxy-domain-sales-o2c

Component Breakdown

`cpi`	Provider Identifier (SAP Cloud Integration)
`dcrp`	Routing Methodology (Domain-Centric Pattern)
`[Domain]`	Primary Business Area (e.g., Sales, Finance)
`[Subprocess]`	Specific Functional Scope (e.g., O2C, R2R)

Strategic Benefit: Subprocess Segregation

Isolating by subprocess (e.g., **o2c**) prevents the "Monolithic Proxy" trap. Updates to Warehouse Management won't risk the stability of the entire Logistics domain.

Domain Examples

Sales
Order to Cash (O2C)
`...-sales-o2c`
`/sales`

Logistics
Supply Chain (SCM)
`...-logistics-scm`
`/logistics`

Finance
Record to Report (R2R)
`...-finance-r2r`
`/finance`

Procurement
Source to Pay (S2P)
`...-procurement-s2p`
`/procurement`

Base URL Contract: 1 per Domain

Consistent Naming

ProxyEndpoint: Routing Rules

The Gatekeeper: Controlling Traffic Entry to the Engine

Route Configuration

Defining the Entry Criteria

ProxyEndpoint XML

```
<RouteRulename="orders"> >
  <Condition>
   proxy.pathsuffix MatchesPath"/orders/**"
  </Condition>
  <TargetEndpoint> default </TargetEndpoint>
</RouteRule>
```

How It Works

MatchesPath acts as the security guard. It validates the address on the the envelope before letting it into the building.

/orders/: The specific "Department"

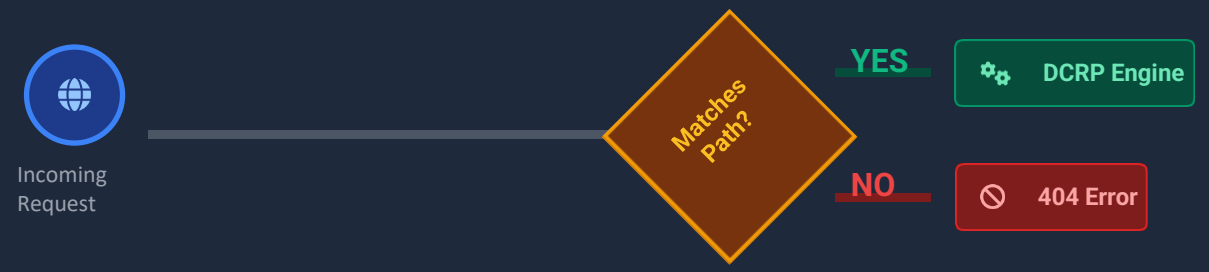
**: The Wildcard ("anything else")

Matching Logic

Path Evaluation Examples

Incoming Path	Match?	Reasoning
/orders/123	✓ YES	Starts with /orders/
/orders/list/today	✓ YES	Wildcard (**) covers depth
/customers/456	✗ NO	Missing "orders" prefix
/orders	✓ YES	Exact folder match

Traffic Flow Visualization



TargetEndpoint & KVM Configuration

Connecting the Proxy to the Backend Metadata

Target Properties

Base Connectivity Settings

cpi.host	`https://[tenant].cfapps.eu20.hana.ondemand.com`
cpi.basepath	`/http/orders`

Lookup Relationship

API Proxy (Sales Domain)

MapIdentifier Lookup

KVM: cpipackage-nx...

KVM Structure

One Map per CPI Package

Map Identifier (Naming Convention)

cpipackage-nx.sales.domain.process

1. packagename (Technical ID)

`nx.sales.domain.process`

2. sapprocess (Business Description)

`SAP SD & O2C - Sales Process`

3. iflowname (Routing Metadata)

```
inbound:SO.01.Iflow.Sales.Order.Inbound,
tracking:SO.02.Iflow.Sales.Order.Tracking
```

Format: `httppath:IflowTechnicalName`

The Semantic Facade

One Proxy, Multiple Intentions: Decoupling Public Contract from Backend Implementation

Architecture Pattern

Abstraction Layer

Consumer View (Public Contract)

INBOUND ORDER — POST

`.../sales/orders/inbound`

TRACKING — GET

`.../sales/orders/tracking`

STATUS CHECK — GET

`.../sales/orders/status`

The Semantic Facade (DCRP Proxy)

Unified Gateway

Sales Domain Proxy

Single Entry Point for all Sales Operations

- OAuth2 Security ✓
- Traffic Control ✓
- Dynamic Routing ✓

DCRP Engine

KVM Lookup
Path → iFlow

JS Routing
Dynamic URL

Backend Reality (CPI iFlows)

iFlow: Inbound
`.../http/orders/inbound`

iFlow: Tracking
`.../http/orders/tracking`

iFlow: Status
`.../http/orders/status`

Total Abstraction
Consumer never sees backend URLs

Unified Monitoring
Full domain visibility in one dashboard

Centralized Security
One OAuth2 policy for all paths

Unified Traffic Control
Quotas & Spike Arrest by domain

Scaling Strategies Comparison

Operational Models: Manual Control vs. Dynamic Velocity

Strategy

Model Selection

Approach 1: Manual

Stable / Small

Explicit routing rules defined individually.

Rule: /orders/**

Rule: /customers/** N Rules

Rule: /...

- 1 Rule per Path
- Fixed cpi.basepath
- ~2 min per new path

Best For:
< 5 Paths or Stable Domains

Approach 2: Dynamic

Enterprise

Catch-all routing with smart JS parsing.

/** JS Parser

- Catch-all + JS Logic
- Comma-delimited string
- ~30 sec per new path

```
cpi.basepath = "/http/orders,/http/customers..."
```

Feature	Approach 1 (Manual)	Approach 2 (Dynamic)
Routing Rule	N rules (1 per path)	**1 catch-all (/**)**
Configuration	Fixed per rule	**Comma-delimited string**
Security	Rule-based	Whitelist Validation
Time / Path	**~2 min (add rule)**	**~30 sec (update string)**

ROI Analysis

62% Faster

Scenario: 4 Domains × 8 Paths

Manual 64 min

Dynamic 24 min

(4 domains × 1 setup) + (32 paths × 30s) = 24 mins

Security Shield v1.0

Defense-in-Depth for Dynamic Catch-All Routes

Cybersecurity

Active Defense

Configuration & Logic

~30s Setup

```
// JS-Security-Shield.js
var allowedDomains = [
"sales", "orders",
"customers", "invoices",
  "payments" // <-- New Path
];
```

Policy Execution Order

1. Proxy PreFlow — Request /**
2. Security Shield — Validate Whitelist
3. Quota/Spike — Rate Limiting
4. Target PreFlow — KVM + Routing Engine

Threat Matrix

Blocked at PreFlow

Attack Vector	Payload Example	Result
Path Traversal	`/sales/../admin/del`	403 Forbidden
SQL Injection	`/orders?id=1 OR 1=1`	403 Forbidden
Domain Hijacking	`/hacker-domain/run`	403 Forbidden
Command Inj.	`/orders?cmd=;cat /etc`	403 Forbidden

~100ms
Saved per Blocked Attack

SOC2 / ISO 27001
Compliance Ready

API Contracts & OpenAPI

Hybrid Model: Immutable Gateway, Dynamic Versioning

Governance

Lifecycle Mgmt

Routing Examples

.../orders/v1/inbound	→	iFlow v1 (Legacy)
.../orders/v4/inbound	→	iFlow v4 (Beta)
.../orders/inbound	→	iFlow v2 (Default)

Key Benefits:

- ✓ No Proxy Sprawl
- ✓ Backwards Compat.

KVM vs OpenAPI

Roles Separation

Aspect	KVM (Metadata)	OpenAPI (Contract)
Core Question	"Where should I route this?"	"What schema is required?"
Lifecycle	Runtime	
Input	Path Suffix (/v2)	Request Body/Headers
Function	Dynamic Dispatcher	Validation & Docs

Validation Metrics

Real-World Results: Efficiency, Speed, and Reliability

Performance
Dashboard

Infrastructure
90%
↓ 40 to 4 Proxies

Deployment
95%
↓ 273m to 14.5m

Volume
33k+
100% Success

Latency
68ms
Average RTT

Domain Deployment Scope

4 Domains Live

Business Domain	API Proxy ID	CPI Package	iFlows	Time
Sales & O2C	cpi-dcrp-sales	nx.sales	2	2 min
Logistics & SCM	cpi-dcrp-logistics	nx.logistics	2	2 min
Finance & R2R	cpi-dcrp-finance	nx.finance	2	2 min
Procurement & S2P	cpi-dcrp-procurement	nx.procurement	2	2 min

Performance

Total iFlows	8
Total Time	~8 min
Error Rate	0%

Resilience Architecture

Fault Isolation:
If one endpoint (e.g., /inbound) is missing, it returns a precise 404 without affecting the routing logic for other paths in the same domain.

Statistics Collector Policy

Capturing Business Intelligence & Operational Metadata

Intelligence
Analytics

Correlation ID	iFlow Name	CPI Package	SAP Process	Sender ID
`idcorrelation`	`idiflow`	`idpackage`	`idsapprocess`	`idsender`

StatisticsCollector.xml

Policy Config

```
<StatisticsCollector async='false' continueOnError='false' enabled='true'>
<Statistics>
<Statistic name='custom_idcorrelation' ref='idcorrelation' type='string'>0</Statistic>
<Statistic name='custom_idiflow' ref='idiflow' type='string'>0</Statistic>
<Statistic name='custom_idpackage' ref='idpackage' type='string'>0</Statistic>
<Statistic name='custom_idsapprocess' ref='idsapprocess' type='string'>0</Statistic>
<Statistic name='custom_idsender' ref='idsender' type='string'>0</Statistic>
</Statistics>
</StatisticsCollector>
```

Strategic Utility

- Visualize traffic spikes by business process.
- Identify high-load iFlows needing scaling.
- Full traceability from Sender to CPI Package.

Configuration Tip

Ensure the ref attribute (e.g., idiflow) matches exactly the variable name set in your JavaScript policy: context.setVariable("idiflow", ...).

JS Engine > Collector > Analytics DB

Troubleshooting Reference

Common Issues, Root Causes, and Rapid Resolutions

Operations

Support Guide

Error Code / Type	Root Cause	Fix / Solution	Severity
Runtime Errors			
404 Not Found	KVM `iflowname` delimiter mismatch (e.g., using space instead of :)	Verify separator is `:` in KVM (e.g., `inbound:SO.01...`). Check if endpoint exists.	High
500 Internal Error	Missing KVM or incorrect `mapIdentifier` in policy configuration.	Ensure KVM map `cpipackage-nx.[domain]` exists and is deployed.	Critical
403 Forbidden	Domain path not whitelisted in Security Shield logic.	Add domain to `allowedDomains` array in JS-Security-Shield.js.	High
504 Gateway Timeout	Incorrect `cpi.host` or slow backend response.	Verify Target Endpoint properties. Check CPI iFlow performance logs.	High
Configuration Issues			
KVM Returns Null	Environment mismatch (e.g., Proxy in Dev, KVM in Prod).	Deploy KVM to the **same environment** as the API Proxy.	Medium
JS Execution Error	Unhandled exception in routing script (e.g., null variable).	Add `try/catch` blocks and default values: `var x = context.get() || "default";`	Medium
Rule Doesn't Match	Routing Rule Condition is too specific for dynamic routing.	Change Condition to `MatchesPath "/**"` for Approach 2 (Dynamic).	Medium
Stats Not Logging	Variable name mismatch in Statistics Collector policy.	Ensure `<Statistic ref='...'>` matches JS `context.setVariable()` exactly.	Low
Security Shield Issues			
All Requests 403	Case sensitivity issue in whitelist array.	Ensure `allowedDomains` values are lowercase (e.g., "sales", not "Sales").	High
SQLi False Positive	Regex too broad, blocking valid IDs with hyphens/special chars.	Refine regex pattern to exclude valid business data formats.	Medium
Performance Issues			
High Latency (>80ms)	JavaScript parsing overhead or network latency to CPI.	Profile with APIM Trace. Optimize KVM string length or check network.	Medium

When to Escalate

SAP Support / Level 3 Criteria

- Issue reproducible in isolated test proxy
- KVM API returns 500 despite valid config
- **Collect:** Bundle, Trace Logs, Network HAR

ROI & Business Case

Quantifying the Value of Governance & Consolidation

Strategy

Cost Efficiency

Deployment Time Formula

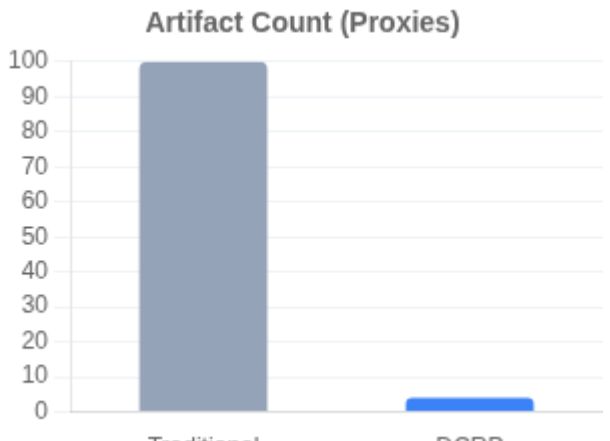
DCRP v1.0

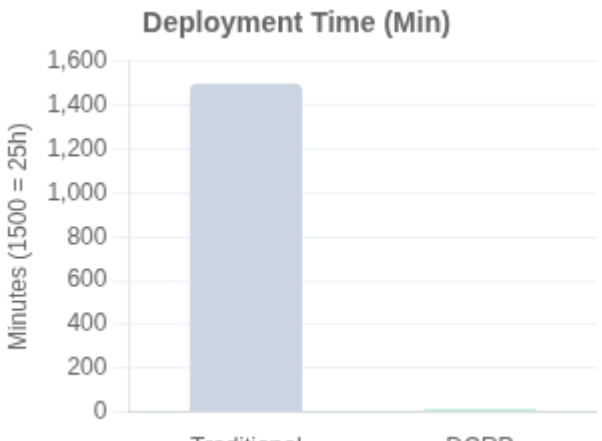
$$\text{Total Time} = (N_{domains} \times 2 \text{ min}) + (M_{iFlows} \times 30 \text{ sec})$$

N: Number of Business Domains **M:** Total Integration Flows

Infrastructure & Speed Comparison

100 Integrations Scenario





Tangible Benefits

- **Operational Cost:** Massive reduction in maintenance overhead for proxy configurations.
- **Rapid Deployment:** New integrations live in seconds, not hours.
- **Standardization:** Replicable templates eliminate configuration drift.

Intangible Benefits

- **Centralized Security:** Single policy enforcement point per domain.
- **Traceability:** Unified analytics across all domain traffic.
- **Lower Risk:** Metadata changes (KVM) are safer than code deployments.

Roadmap & Conclusions

Strategic Path to Enterprise Integration Maturity

Next Steps

Future View

You Are Here

Part I: DCRP Gateway

Solving Proxy Sprawl
Metadata-Driven Routing

KVM + JavaScript

Part II: PDCP Orch.

Solving Package Sprawl
Domain Consolidation

Coming Soon

Part III: Implementation

Hands-on Guide
JS v8.0, GitHub Repo

Coming Soon

Pre-Flight Checklist

Before scaling DCRP to production

- **Team Alignment:** Devs understand 1:1 shift
- **Education:** Stakeholders grasp value
- **Sandbox PoC:** Validate with 2-3 iFlows first
- **Governance:** KVM update workflows
- **Standards:** Naming conventions fixed

Intellectual Property & Usage

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